Reflection Analysis

Input Parameters

Generel

Data file:	*26-10-3 Emerged Berm 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *35.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1622*	*0.05787*
Peak wave period T_P (s):	*1.73*	*1.73*
Energy wave period $T_{-1,0}$ (s):	*1.546*	*1.433*
Mean wave period $T_{0,1}$ (s):	*1.393*	*1.022*
Mean wave period $T_{0,2}$ (s):	*1.286*	*0.8515*
Spectral width (Broadness):	*0.8075*	*0.8302*
Spectral width (Narrowness):	*0.4167*	*0.6645*

Time Domain Analysis

Overall amplitude reflection coefficient: *35.6%*

	Incident	**Reflected**
Number of waves:	959	1203
Mean wave height H_m (m):	*0.1105*	*0.03939*
Mean wave period T_m (s):	*1.488*	*1.187*
Sig. wave height H_s (m):	*0.1591*	*0.05913*
$T_{H_{1/3}}$ (s):	*1.63*	*1.513*
H_{max} (m):	*0.2285*	*0.09509*
T_{Hmax} (s):	*1.461*	*1.453*
$H_{1/10}$ (m):	*0.1822*	*0.07084*
$H_{1/50}$ (m):	*0.2047*	*0.08139*
$H_{1/100}$ (m)	*0.2113*	*0.08537*
$H_{1/250}$ (m):	*0.2178*	*0.09072*
$H_{0.1\%}$ (m):	*Not enough data*	*0.09495*
$H_{1\%}$ (m):	*0.2038*	*0.07929*
$H_{2\%}$ (m):	*0.1912*	*0.07631*
$H_{10\%}$ (m)	*0.1663*	*0.06252*
Groupiness factor GF:	*0.8465*	*0.9925*
Skewness b1:	*0.5738*	*0.1573*
Kurtosis b2:	*2.663*	*2.996*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*11. mrt 2026*
Remarks	
Page	*1 of 8*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment - 100% of 4 hours*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.359 Hz*
Freq. upper bound:	*7.188 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*23*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*27*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1246*	*0.03787*
Peak wave period T_P (s):	*1.391*	*1.488*
Energy wave period $T_{-1,0}$ (s):	*1.338*	*1.159*
Mean wave period $T_{0,1}$ (s):	*1.252*	*0.9162*
Mean wave period $T_{0,2}$ (s):	*1.191*	*0.8023*
Spectral width (Broadness):	*0.7272*	*0.8033*
Spectral width (Narrowness):	*0.3248*	*0.5514*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.4%*

	Incident	**Reflected**
Number of waves:	948	1205
Mean wave height H_m (m):	*0.07975*	*0.0248*
Mean wave period T_m (s):	*1.252*	*0.9853*
Sig. wave height H_s (m):	*0.1231*	*0.03803*
$T_{H_{1/3}}$ (s):	*1.389*	*1.162*
H_{max} (m):	*0.1825*	*0.06684*
T_{Hmax} (s):	*1.449*	*1.939*
$H_{1/10}$ (m):	*0.1493*	*0.04732*
$H_{1/50}$ (m):	*0.1696*	*0.05701*
$H_{1/100}$ (m)	*0.1752*	*0.06006*
$H_{1/250}$ (m):	*0.1791*	*0.06373*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06623*
$H_{1\%}$ (m):	*0.17*	*0.05583*
$H_{2\%}$ (m):	*0.1619*	*0.05143*
$H_{10\%}$ (m)	*0.1309*	*0.04071*
Groupiness factor GF:	*0.9485*	*1.064*
Skewness b1:	*0.4244*	*0.2678*
Kurtosis b2:	*2.911*	*3.198*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.352 Hz*
Freq. upper bound:	*7.031 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*24*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*27*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *32%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1384*	*0.04435*
Peak wave period T_P (s):	*1.422*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.387*	*1.21*
Mean wave period $T_{0,1}$ (s):	*1.285*	*0.9271*
Mean wave period $T_{0,2}$ (s):	*1.211*	*0.8007*
Spectral width (Broadness):	*0.7651*	*0.8163*
Spectral width (Narrowness):	*0.3553*	*0.5839*

Time Domain Analysis

Overall amplitude reflection coefficient: *32.1%*

	Incident	Reflected
Number of waves:	944	1262
Mean wave height H_m (m):	*0.09008*	*0.02854*
Mean wave period T_m (s):	*1.321*	*0.9888*
Sig. wave height H_s (m):	*0.1355*	*0.04402*
$T_{H_{1/3}}$ (s):	*1.466*	*1.207*
H_{max} (m):	*0.2203*	*0.08837*
T_{Hmax} (s):	*1.512*	*0.5191*
$H_{1/10}$ (m):	*0.1606*	*0.05622*
$H_{1/50}$ (m):	*0.1855*	*0.0698*
$H_{1/100}$ (m)	*0.1946*	*0.07478*
$H_{1/250}$ (m):	*0.2035*	*0.0804*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08724*
$H_{1\%}$ (m):	*0.1841*	*0.06792*
$H_{2\%}$ (m):	*0.1698*	*0.06194*
$H_{10\%}$ (m)	*0.1424*	*0.04719*
Groupiness factor GF:	*0.9211*	*1.107*
Skewness b1:	*0.5155*	*0.2224*
Kurtosis b2:	*2.859*	*3.374*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *32.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1456*	*0.04762*
Peak wave period T_P (s):	*1.524*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.445*	*1.291*
Mean wave period $T_{0,1}$ (s):	*1.33*	*0.9705*
Mean wave period $T_{0,2}$ (s):	*1.248*	*0.8299*
Spectral width (Broadness):	*0.7696*	*0.8184*
Spectral width (Narrowness):	*0.3674*	*0.6063*

Time Domain Analysis

Overall amplitude reflection coefficient: *32.7%*

	Incident	**Reflected**
Number of waves:	949	1260
Mean wave height H_m (m):	*0.09664*	*0.03118*
Mean wave period T_m (s):	*1.382*	*1.041*
Sig. wave height H_s (m):	*0.144*	*0.0478*
$T_{H_{1/3}}$ (s):	*1.525*	*1.29*
H_{max} (m):	*0.2237*	*0.08724*
T_{Hmax} (s):	*1.572*	*1.901*
$H_{1/10}$ (m):	*0.1692*	*0.06007*
$H_{1/50}$ (m):	*0.1901*	*0.07112*
$H_{1/100}$ (m)	*0.1961*	*0.07442*
$H_{1/250}$ (m):	*0.2057*	*0.07783*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08476*
$H_{1\%}$ (m):	*0.1884*	*0.0709*
$H_{2\%}$ (m):	*0.1811*	*0.06523*
$H_{10\%}$ (m)	*0.1528*	*0.05154*
Groupiness factor GF:	*0.8983*	*1.046*
Skewness b1:	*0.5235*	*0.1629*
Kurtosis b2:	*2.784*	*3.181*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.3%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1546*	*0.05306*
Peak wave period T_P (s):	*1.524*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.49*	*1.352*
Mean wave period $T_{0,1}$ (s):	*1.358*	*0.9964*
Mean wave period $T_{0,2}$ (s):	*1.264*	*0.8409*
Spectral width (Broadness):	*0.7965*	*0.8328*
Spectral width (Narrowness):	*0.3921*	*0.6358*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.4%*

	Incident	**Reflected**
Number of waves:	957	1300
Mean wave height H_m (m):	*0.1038*	*0.03486*
Mean wave period T_m (s):	*1.429*	*1.052*
Sig. wave height H_s (m):	*0.1527*	*0.05325*
$T_{H_{1/3}}$ (s):	*1.579*	*1.362*
H_{max} (m):	*0.2375*	*0.08765*
T_{Hmax} (s):	*1.506*	*1.058*
$H_{1/10}$ (m):	*0.1771*	*0.06532*
$H_{1/50}$ (m):	*0.2019*	*0.07628*
$H_{1/100}$ (m)	*0.2119*	*0.08022*
$H_{1/250}$ (m):	*0.2249*	*0.08422*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08692*
$H_{1\%}$ (m):	*0.1979*	*0.07477*
$H_{2\%}$ (m):	*0.1892*	*0.07058*
$H_{10\%}$ (m)	*0.161*	*0.05704*
Groupiness factor GF:	*0.8769*	*0.9925*
Skewness b1:	*0.5779*	*0.1841*
Kurtosis b2:	*2.756*	*2.998*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment - 150% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1678*	*0.06216*
Peak wave period T_P (s):	*1.73*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.582*	*1.493*
Mean wave period $T_{0,1}$ (s):	*1.411*	*1.052*
Mean wave period $T_{0,2}$ (s):	*1.293*	*0.8673*
Spectral width (Broadness):	*0.8236*	*0.8411*
Spectral width (Narrowness):	*0.437*	*0.6861*

Time Domain Analysis

Overall amplitude reflection coefficient: *37%*

	Incident	**Reflected**
Number of waves:	931	1168
Mean wave height H_m (m):	*0.1159*	*0.04236*
Mean wave period T_m (s):	*1.533*	*1.222*
Sig. wave height H_s (m):	*0.164*	*0.06324*
$T_{H_{1/3}}$ (s):	*1.673*	*1.621*
H_{max} (m):	*0.2216*	*0.1067*
T_{Hmax} (s):	*1.504*	*1.221*
$H_{1/10}$ (m):	*0.187*	*0.07619*
$H_{1/50}$ (m):	*0.2104*	*0.09061*
$H_{1/100}$ (m)	*0.2163*	*0.09536*
$H_{1/250}$ (m):	*0.2189*	*0.1007*
$H_{0.1\%}$ (m):	*Not enough data*	*0.1057*
$H_{1\%}$ (m):	*0.2131*	*0.08878*
$H_{2\%}$ (m):	*0.1973*	*0.08368*
$H_{10\%}$ (m)	*0.172*	*0.06711*
Groupiness factor GF:	*0.8382*	*0.9794*
Skewness b1:	*0.6079*	*0.1285*
Kurtosis b2:	*2.651*	*2.891*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment data (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.461 Hz*
Freq. upper bound:	*9.219 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*19*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*21*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *26.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.07249*	*0.01911*
Peak wave period T_P (s):	*1.085*	*1.085*
Energy wave period $T_{-1,0}$ (s):	*1.058*	*0.9554*
Mean wave period $T_{0,1}$ (s):	*1.025*	*0.8331*
Mean wave period $T_{0,2}$ (s):	*1.005*	*0.7639*
Spectral width (Broadness):	*0.4914*	*0.7293*
Spectral width (Narrowness):	*0.2029*	*0.435*

Time Domain Analysis

Overall amplitude reflection coefficient: *26.3%*

	Incident	**Reflected**
Number of waves:	986	1182
Mean wave height H_m (m):	*0.04553*	*0.01229*
Mean wave period T_m (s):	*1.022*	*0.8525*
Sig. wave height H_s (m):	*0.07118*	*0.01859*
$T_{H_{1/3}}$ (s):	*1.053*	*0.9731*
H_{max} (m):	*0.1284*	*0.03289*
T_{Hmax} (s):	*1.05*	*0.6895*
$H_{1/10}$ (m):	*0.09058*	*0.02294*
$H_{1/50}$ (m):	*0.109*	*0.02775*
$H_{1/100}$ (m)	*0.1145*	*0.02953*
$H_{1/250}$ (m):	*0.124*	*0.03147*
$H_{0.1\%}$ (m):	*Not enough data*	*0.03275*
$H_{1\%}$ (m):	*0.1064*	*0.02771*
$H_{2\%}$ (m):	*0.1015*	*0.02497*
$H_{10\%}$ (m)	*0.07584*	*0.01962*
Groupiness factor GF:	*0.984*	*0.9715*
Skewness b1:	*0.0386*	*0.1252*
Kurtosis b2:	*2.997*	*2.909*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-10 Emerged Berm 4% - test 1.experiment - 80% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.398 Hz*
Freq. upper bound:	*7.969 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*21*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*24*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *28.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1035*	*0.02974*
Peak wave period T_P (s):	*1.255*	*1.255*
Energy wave period $T_{-1,0}$ (s):	*1.214*	*1.022*
Mean wave period $T_{0,1}$ (s):	*1.16*	*0.8414*
Mean wave period $T_{0,2}$ (s):	*1.121*	*0.7558*
Spectral width (Broadness):	*0.6482*	*0.7603*
Spectral width (Narrowness):	*0.2646*	*0.4893*

Time Domain Analysis

Overall amplitude reflection coefficient: *28.7%*

	Incident	**Reflected**
Number of waves:	977	1255
Mean wave height H_m (m):	*0.06661*	*0.01948*
Mean wave period T_m (s):	*1.154*	*0.8982*
Sig. wave height H_s (m):	*0.1028*	*0.02983*
$T_{H_{1/3}}$ (s):	*1.233*	*1.017*
H_{max} (m):	*0.1772*	*0.05849*
T_{Hmax} (s):	*1.244*	*1.713*
$H_{1/10}$ (m):	*0.126*	*0.03754*
$H_{1/50}$ (m):	*0.1471*	*0.0461*
$H_{1/100}$ (m)	*0.1537*	*0.04932*
$H_{1/250}$ (m):	*0.1629*	*0.05369*
$H_{0.1\%}$ (m):	*Not enough data*	*0.05778*
$H_{1\%}$ (m):	*0.1447*	*0.04519*
$H_{2\%}$ (m):	*0.1386*	*0.04161*
$H_{10\%}$ (m)	*0.1099*	*0.03146*
Groupiness factor GF:	*0.9357*	*1.093*
Skewness b1:	*0.1657*	*0.09546*
Kurtosis b2:	*2.896*	*3.267*